Reproduction of Table I of arXiv:1708.02243

Knapen, Lou, Papucci, Setford, Phys. Rev. D 96, 115015 (2017) · 15 October 2026 · production samples 2000 events each · EW LO sample: first 2495 of 3000 events (run in progress)

Questions to answer

- **Q1. Why is the converted-photon upper limit the same for both cross sections?** → Section 7.
- **Q2. Figure 2: why does our |L| go to zero at every second meeting while the paper's does not?** → Section 3 (with Appendix A of arXiv:2002.07503).
- **Q3. Which smearing was applied, and how?** → Section 1.
- **Q4. Which uncertainties are quoted?** → Section 1 (formula) and every table (k/n = ε ± σ).

1. How the hits are smeared, and the uncertainty used

Detector part	Formula applied to the true crossing point	Value
Pixel layers (4)	hit moved to the centre of its pixel: $x_{meas} = p \cdot \mathrm{round}(x / p)$	p = 50 μm (rφ); 250 μm (IBL z), 400 μm (other z)
SCT layers (4), rφ	$(r\varphi)_{meas} = r\varphi + \sigma_{r\varphi} \cdot g$, g = Gaussian random number (mean 0, width 1)	$\sigma_{r\varphi}$ = 34 μm (2 × 17 μm)
SCT layers (4), z	$z_{meas} = z + \sigma_z \cdot g$	σ_z = 0 (“no smearing”) or 1.16 mm (2 × 580 μm)
Beam spot	all hits of an event shifted by the same z_0, z_0 = 45 mm · g	45 mm
Neighbouring pixels	hits in adjacent pixels of one layer merged into one hit	—

Pixel rounding gives an RMS error p/√12 (14 μm for 50 μm, 115 μm for 400 μm). Everything is from the paper (Sec. III A) and Ref. [27], except σ_z = 0 (our choice: it is the only value that reproduces the paper's numbers) and one common z_0 per event (our reading).

Uncertainty of an efficiency ε = k / n: $\sigma = \sqrt{\varepsilon\,(1-\varepsilon)\,/\,n}$ (statistical only)

Decision: **consistent** with the paper if $|\varepsilon - \varepsilon_{paper}| \le 2\,\sigma_0$, $\sigma_0 = \sqrt{\varepsilon_{paper}\,(1-\varepsilon_{paper})\,/\,n}$

2. Table I

Colored quirk, m = 1800 GeV (2000 events generated, 873 after matching)

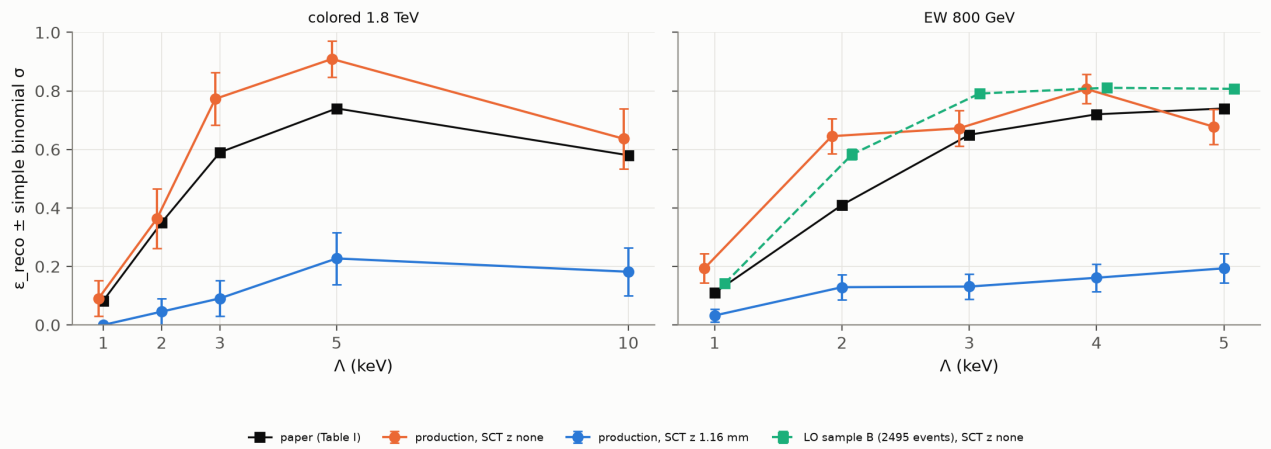
Λ (keV)	ε_{trig} = N(p_T(QQ) > 200 GeV) / N(matched)	ε_{fid} = N(fiducial) / N(trig., both charged)	ε_{reco} = N(signal region) / N(fiducial), no smearing	paper ε_{reco}	decision	ε_{reco}, SCT z 1.16 mm	decision
1	207/873 = 0.24 ± 0.01	22/68 = 0.32 ± 0.06	2/22 = 0.09 ± 0.06	0.083	consistent	0/22 = 0.00 ± 0.00	consistent
2	same	22/68 = 0.32 ± 0.06	8/22 = 0.36 ± 0.10	0.35	consistent	1/22 = 0.05 ± 0.04	**inconsistent**
3	same	22/68 = 0.32 ± 0.06	17/22 = 0.77 ± 0.09	0.59	consistent	2/22 = 0.09 ± 0.06	**inconsistent**
5	same	22/68 = 0.32 ± 0.06	20/22 = 0.91 ± 0.06	0.74	consistent	5/22 = 0.23 ± 0.09	**inconsistent**
10	same	22/68 = 0.32 ± 0.06	14/22 = 0.64 ± 0.10	0.58	consistent	4/22 = 0.18 ± 0.08	**inconsistent**
paper	0.24 → consistent	0.28 → consistent					

EW quirk, m = 800 GeV (2000 events generated, 1045 after matching)

Λ (keV)	ε_{trig} = N(p_T(QQ) > 200 GeV) / N(matched)	ε_{fid} = N(fiducial) / N(trig.)	ε_{reco} = N(signal region) / N(fiducial), no smearing	paper ε_{reco}	decision	ε_{reco}, SCT z 1.16 mm	decision
1	186/1045 = 0.18 ± 0.01	62/186 = 0.33 ± 0.03	12/62 = 0.19 ± 0.05	0.11	**inconsistent**	2/62 = 0.03 ± 0.02	consistent
2	same	62/186 = 0.33 ± 0.03	40/62 = 0.65 ± 0.06	0.41	**inconsistent**	8/62 = 0.13 ± 0.04	**inconsistent**
3	same	61/186 = 0.33 ± 0.03	41/61 = 0.67 ± 0.06	0.65	consistent	8/61 = 0.13 ± 0.04	**inconsistent**
4	same	62/186 = 0.33 ± 0.03	50/62 = 0.81 ± 0.05	0.72	consistent	10/62 = 0.16 ± 0.05	**inconsistent**
5	same	62/186 = 0.33 ± 0.03	42/62 = 0.68 ± 0.06	0.74	consistent	12/62 = 0.19 ± 0.05	**inconsistent**
paper	0.10 → inconsistent	0.28 → consistent					

EW sample	ε_{trig}	ε_{fid} (4 keV)	ε_{reco} at 1 / 2 / 3 / 4 / 5 keV (no smearing)
Production: MadGraph + Pythia, matched, showered	0.178 ± 0.012	0.333 ± 0.035	0.19 / 0.65 / 0.67 / 0.81 / 0.68
LO, no shower: $Q\bar{Q}$ + 1 jet with $p_T(j) > 200$ GeV (2495 events)	0.096 ± 0.001 (= $\sigma(Q\bar{Q}j)/\sigma(Q\bar{Q})$; MG5 2.3.3: 0.096)	0.298 ± 0.009	0.14 / 0.58 / 0.79 / 0.81 / 0.81
Paper	0.10	0.28	0.11 / 0.41 / 0.65 / 0.72 / 0.74

With the LO sample the EW trigger (0.096) and fiducial (0.30) efficiencies match the paper; the reconstruction efficiency does not change. For colored, the same LO ratio gives 0.14 (paper 0.24), and the matched sample matches (0.24).



What we learn. Colored: no smearing consistent at every Λ; 1.16 mm inconsistent at Λ = 2, 3, 5, 10 keV. EW: no smearing inconsistent at Λ = 1, 2 keV (higher than the paper), consistent elsewhere; 1.16 mm inconsistent at Λ = 2, 3, 4, 5 keV. The SCT z smearing decides the result: without it our ε_{reco} is close to the paper, with the paper's 1.16 mm it is far below.

3. Figure 2 and Appendix A of arXiv:2002.07503

Equations used (the paper's model, solved numerically):

Eq. 3: $d(m\gamma\mathbf{v})/dt = -\Lambda^2 [\sqrt{(1-v_\perp^2)}\,\hat{s} + (v_\parallel/\sqrt{(1-v_\perp^2)})\,\mathbf{v}_\perp] + q\,\mathbf{v}\times\mathbf{B}$ (string + magnetic force)

$L = (\mathbf{x}_1 - \mathbf{x}_2)\times(\mathbf{p}_1 - \mathbf{p}_2)/2$ in the pair's rest frame; $d = |\mathbf{x}_1 - \mathbf{x}_2|$; r = distance of the pair's centre from the beam

Eq. 5: in the pair's rest frame the magnetic field acts as an electric field $\mathbf{E}^* = \gamma_{cm}\,\mathbf{v}_{cm}\times\mathbf{B}$; torque $\boldsymbol{\tau} = q\,\mathbf{d}\times\mathbf{E}^*$

What Appendix A of arXiv:2002.07503 shows (Li, Li, Pei, Zhang, "The quirk trajectory", same equation, uniform magnetic field):

- Eqs. (II.11)–(II.14): in the pair's rest frame the two quirks move back and forth on one line and meet at every integer g ($g = \Lambda^2 t/(2m\rho)$ is the time in units of half an oscillation).
- Text after Eq. (A.4): "the average torque on the system of two quirks is zero when g is increased from 2n to 2n + 2 such that the total angular momentum of the quirk pair is also invariant at g = 2n".
- Eq. (A.5): $z_1(2n) = -z_2(2n) = 0$, i.e. at every second meeting the quirks meet head-on, as at production.

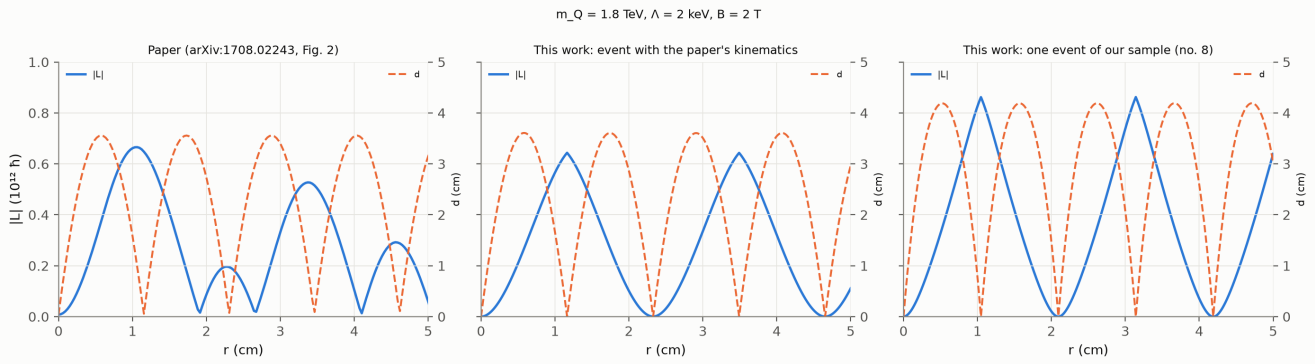
Why. Between two meetings the separation **d** points one way, in the next half-oscillation the other way. The field **E*** is constant, so the torque $q\mathbf{d}\times\mathbf{E}^*$ has the same size and the opposite sign in the two halves: L goes up, then back to zero. Our simulation shows exactly this.

Change of L in each half-oscillation (event with the paper's kinematics, units 10^{12} ħ)	1st	2nd	3rd	4th
total change of L in our simulation	+0.64	-0.64	+0.64	-0.64
from the magnetic force (torque $q\mathbf{d}\times\mathbf{E}^*$)	+0.56	-0.56	+0.56	-0.56
from the string's own term in Eq. 3	+0.09	-0.09	+0.09	-0.09
paper's Fig. 2 (read from its curves)	**+0.65**	**-0.84**	**+0.71**	**-0.80**

In our simulation each half-oscillation adds or removes the same amount (±0.64), so L returns to zero after every two halves. In the paper the halves are unequal (+0.65, −0.84, +0.71, −0.80), so L never returns to zero at a meeting.

Fig. 2 value	Paper (arXiv:1708.02243)	This work: event with the paper's kinematics[1]	This work: one event of our sample[2]
largest separation d (cm)	3.56	3.61	4.19
distance in r between meetings (cm)	1.15	1.16	1.05
|L| at meetings 1–4 (10^{12} ħ)	0.65 / 0.19 / 0.51 / 0.29	0.64 / 0.00 / 0.64 / 0.00	0.86 / 0.00 / 0.86 / 0.00
match with the paper	—	d and first |L|: yes; later |L|: no	pattern as in the middle column

[1] $v^* = 0.556$ and centre speed 0.0975 c, read from the paper's d curve; charges ±2/3. [2] Event no. 8 of our colored sample (MadGraph + Pythia), chosen because its d curve is the closest to the paper's; charges ±1.



Conclusion. Our Fig. 2 is what the paper's own equation gives; the same result is found analytically in arXiv:2002.07503 (Appendix A) and with an independent code. The paper's later |L| values cannot be obtained from its stated equation; the paper does not give the event's charges and momenta or how L was computed.

4. Does L = 0 mean the quirks annihilate?

- The paper says |L| = 0 only when the quirks are far apart, so they cannot annihilate. In our simulation (and in Appendix A) |L| ≈ 0 every second time they meet. So the paper's reason does not hold.
- L ≈ 0 is necessary but not enough: the quirks must also actually meet. Our classical simulation cannot resolve this; the left-over |L| at these meetings is 10^6–10^8 ħ and keeps shrinking with smaller time steps.
- Worst case (annihilation possible at every second meeting, probability ≈ α^2 γv per oscillation, α = 0.02 or 0.05): fraction of pairs surviving inside the tracker 0.995 / 0.970 (colored, 2 keV), 0.971 / 0.835 (EW, 4 keV), 0.887 / 0.494 (colored, 10 keV). Small at the Fig. 2 point, not small at large Λ.
- Whether this gives non-negligible annihilation needs a quantum estimate of the per-meeting probability, which neither the paper nor this study provides. Table I assumes none.

5. Quirks crossing the same layer more than once

Sample (fiducial events, all crossings within 25 ns)	events with > 2 hits on at least one layer	fraction ± σ
Colored 1800 GeV, 2 keV	13 / 22	0.59 ± 0.10
EW 800 GeV, 4 keV	31 / 62	0.50 ± 0.06

These are results of our simulation. The paper says each quirk “will most likely intersect each layer only once”. arXiv:2002.07503 (Sec. VI, Table I and Appendix D, Table II) also finds repeated crossings, more often at larger Λ and larger m/p_T, with 4–8 hits on some layers at Λ = 2 keV; its percentages (~10%) are for other masses and the CMS tracker and are not directly comparable. Table I above uses the first crossing only.

6. Figures 3, 4, 5

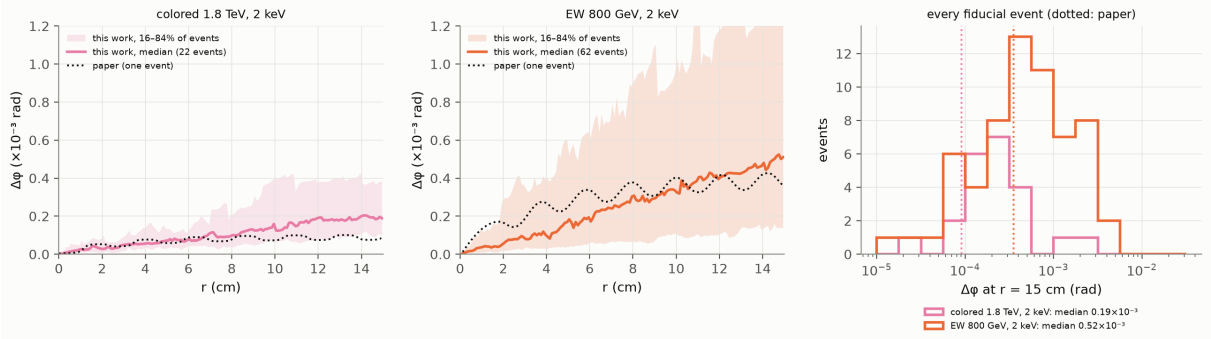


Fig. 3 (turning of the quirks' plane): our spread contains the paper's event; the plane turns by less than 10^{-3} rad. Match: yes.

Fig. 4 example event	Δs (plane thickness)	Δw (strip width)	match with the paper
Paper (colored)	83 µm	0.074 cm	—
This work, colored, no smearing	38 µm	0.352 cm	Δs yes (both below the 100 µm cut); Δw no (wider)
This work, EW, no smearing	45 µm	0.100 cm	Δs yes (both below the 100 µm cut); Δw close
This work, colored, 1.16 mm	52 µm	0.470 cm	Δs yes (both below the 100 µm cut); Δw no (wider)
This work, EW, 1.16 mm	73 µm	0.079 cm	Δs yes (both below the 100 µm cut); Δw close

Each example is the signal event with the median Δs. All hits lie on a thin strip, as in the paper.

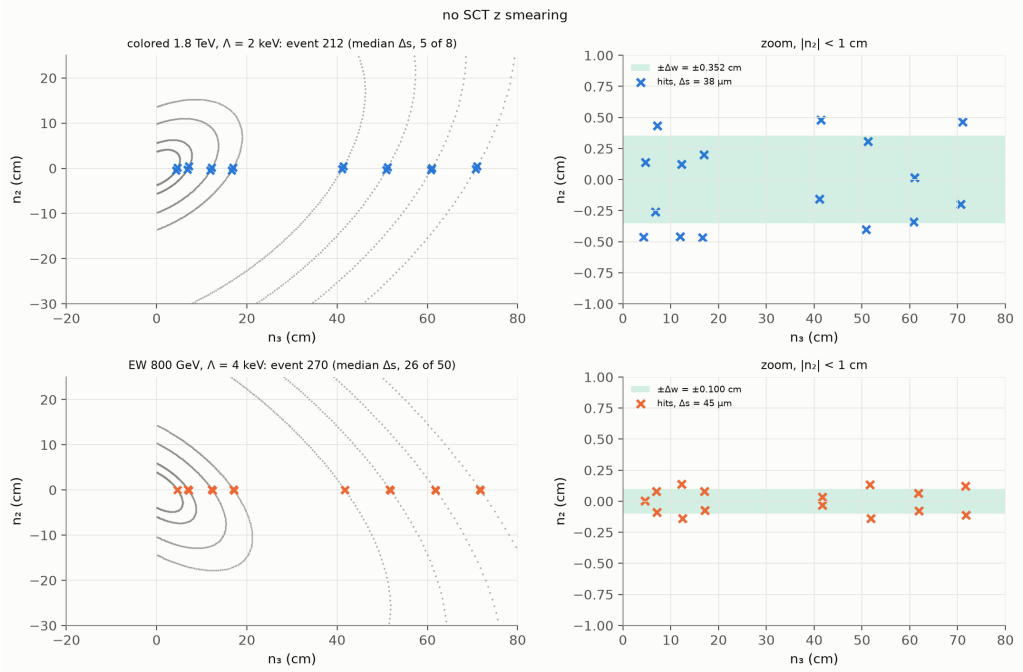
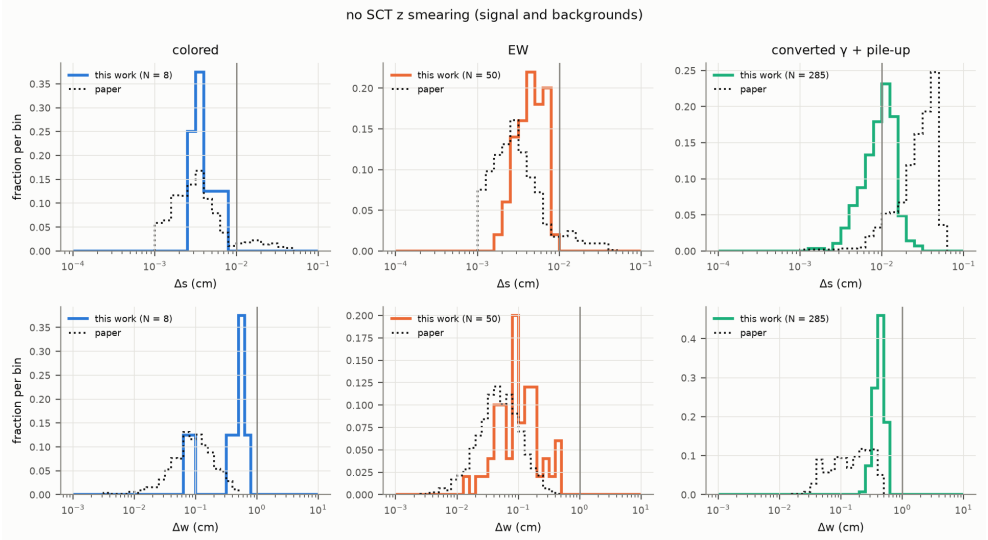


Fig. 5	smearing	Δs median: ours / paper	Δw median: ours / paper	fraction with Δs < 0.01 cm: ours / paper	decision
colored signal	none	38 / 32 μm	0.482 / 0.092 cm	8/8 = 1.00 ± 0.00 / 0.90	consistent
colored signal	1.16 mm	160 / 32 μm	0.550 / 0.092 cm	1/4 = 0.25 ± 0.22 / 0.90	**inconsistent**
EW signal	none	44 / 28 μm	0.097 / 0.048 cm	50/50 = 1.00 ± 0.00 / 0.92	**inconsistent**
EW signal	1.16 mm	110 / 28 μm	0.092 / 0.048 cm	10/24 = 0.42 ± 0.10 / 0.92	**inconsistent**
converted γ + pile-up	none	97 / 306 μm	0.438 / 0.156 cm	146/285 = 0.51 ± 0.03 / 0.07	**inconsistent**
converted γ + pile-up	1.16 mm	222 / 306 μm	0.350 / 0.156 cm	0/48 = 0.00 ± 0.00 / 0.07	consistent
pile-up	1.16 mm	317 / 571 μm	0.778 / 0.525 cm	8/109 = 0.07 ± 0.02 / 0.00	**inconsistent**



What we learn. Without smearing the signal Δs (≈ 40 μm) is close to the paper's (≈ 30 μm) and all signal passes the Δs cut (paper ≈ 90%), but 0.51 of the converted-photon planes also pass (paper 0.07). With 1.16 mm no converted photon passes, as in the paper, but most signal fails too. No single SCT z setting reproduces both the paper's signal and its background. Δw is about 5× (colored) and 2× (EW) the paper's; it follows the width expected from our own kinematics (arXiv:2002.07503 Eq. III.1: Δw ≈ belt width/√12).

7. Converted photons: why the upper limit is the same (Q1)

Converted γ + pile-up (9139 events)	in signal region	per conversion	expected events at 300 fb⁻¹: our σ (0.211 pb) / paper σ (~1 pb)
SCT z 1.16 mm	0 / 9139	< 3.3e-04 (95% UL)	< 0.21 / < 0.98
no SCT z smearing	146 / 9139	0.0160 ± 0.0013	10 / 48

Answer. The limit is a property of our 9139 simulated conversions: 0 pass, so the 95% upper limit per conversion is $1 - 0.05^{1/9139} = 3.3\text{e-}04$, whatever the cross section. The cross section only sets how many

conversions are expected (300 $fb^{-1} \times \sigma \times 1\% = 633$ or 3000), which is why the last column differs. Without SCT z smearing on the background, 146 conversions pass and the background is no longer zero.

8. Open points

- SCT z: the authors' stated resolution gives efficiencies far below Table I; only no smearing reproduces it.
- EW ε_{reco} at $\Lambda = 1$–2 keV above the paper (production and LO sample).
- Signal Δw 2–5× the paper's.
- Paper's Fig. 2: unequal half-oscillations; L definition, charges and momenta of its event not given.
- EW LO sample: 2495/3000 events done; showered version pending.

References

Paper: S. Knapen, H. K. Lou, M. Papucci, J. Setford, Phys. Rev. D 96, 115015 (2017), arXiv:1708.02243. [27] ATLAS, ATL-PHYS-PUB-2009-002.
J. Li, T. Li, J. Pei, W. Zhang, “The quirk trajectory”, arXiv:2002.07503 (Eqs. II.11–II.14, A.4–A.5; Sec. VI, App. D).
J. Li, T. Li, J. Pei, W. Zhang, arXiv:1911.02223 (EW trigger 0.1 as an LO ratio, App. A). J. Li et al., arXiv:2108.06748. J. L. Feng et al., arXiv:2404.13814.
Data and scripts: R_data_2026_10_15.py (all numbers), R_report_2026-10-15_v2.py, F38/F39 figures; decision log D17–D42.